PROJECT 07: INVENTORY MANAGEMENT SYSTEM

DATCOM Lab
NEU-College of Technology
National Economics University
Email: hung.tran@neu.edu.vn

Project Objective

The objective of this project is to design and implement a system to efficiently manage inventory operations, track product quantities across warehouses, manage suppliers, and monitor stock movements and purchase history.

Database and Programming Languages

- **Database Management System:** MySQL
- **Programming Language:** Python

Detailed Project Requirements

System Analysis and Requirements

- Manage detailed information on products, suppliers, warehouses, stock entries, and inventory history.

Main Functionalities

- Product management (add, update, track product information and stock).
- Supplier management (store supplier contacts and information).
- Warehouse management (location, capacity, stock details).
- Manage stock entries and purchase orders.
- Track inventory history (in/out transactions per product and warehouse).
- Generate stock level alerts and inventory reports.

Database Design and Implementation

1. **Data Model Design**
   - Design the Entity-Relationship (ER) diagram.
   - Convert ER diagram to a relational schema with PKs, FKs, and constraints.
2. **Table Structures**
   - Products (ProductID, ProductName, Description, UnitPrice, SupplierID)
   - Suppliers (SupplierID, SupplierName, Address, PhoneNumber)
   - Warehouses (WarehouseID, WarehouseName, Address)
   - StockEntries (EntryID, ProductID, Quantity, EntryDate)
   - InventoryHistory (HistoryID, ProductID, WarehouseID, Quantity, TransactionDate)
3. **Sample Data**
   - Insert 510 records for each table.
   - Generate a database schema diagram using MySQL Workbench.

Advanced Database Objects

- **Indexes:** Improve search speed for frequently queried fields (e.g., ProductName, WarehouseID).
- **Views:** Create views to summarize stock levels per warehouse or supplier delivery history.
- **Stored Procedures:** Automate product restocking, inventory calculations.
- **User Defined Functions:** Calculate stock turnover rates or average delivery time.
- **Triggers:** Update inventory levels upon stock addition or removal.

Database Security and Administration

- Manage user roles (inventory manager, admin) and set access permissions.
- Implement security measures to protect supplier and product data.
- Plan and test regular data backups and recovery operations.
- Use indexing and query optimization to improve system performance.

Python Application Development

- **Database Connection:** Use `mysql-connector-python` or `SQLAlchemy`.
- **Data Management:** Write Python functions to handle product entry, updates, and warehouse operations.
- **Querying and Reporting:** Automate the generation of low-stock alerts and inventory balance reports.
- **Interactive Interface:** Create a command-line tool or basic GUI for interacting with the inventory system.

Deliverables

- A printed comprehensive report including database design, implementation details, and screenshots. Submit your report into LMS. First pages of your report must attach this PDF file.
- Link to publish Github source code and video presenting on Youtube
- SQL schema, sample data scripts, ER diagrams, and Python code.
- Demonstration of functionalities such as adding income, managing expenses, and generating reports.

Conclusion and Recommendations

- Evaluate the efficiency of the implemented system.
- Suggest enhancements such as barcode scanning, integration with ERP, or predictive restocking.

References

- List all documents, tutorials, libraries, and references used during the project.